Understanding Anxiety

A Plain-Language Data Analysis Report

Exploring what daily habits and health signs are linked to feeling anxious

Our Team

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Step 1 — Preparing the Data (Pre-Processing)

Before we can draw any charts, the raw data has to be cleaned. Real-world data is almost never tidy: some answers are missing, some are impossible (like a negative number of sleep hours), and some are written in many different ways. Cleaning the data first makes sure every chart that follows tells the truth.

We started with a survey-style dataset about people's lifestyles and mental health, with about 2,030 people and 19 original questions each — things like age, sleep, caffeine, exercise, heart rate, stress, and whether the person feels anxious.

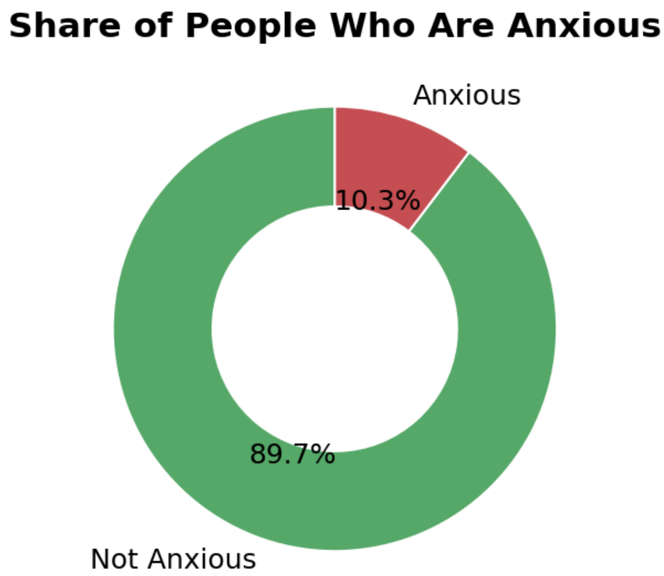
WHAT WE DID TO CLEAN IT

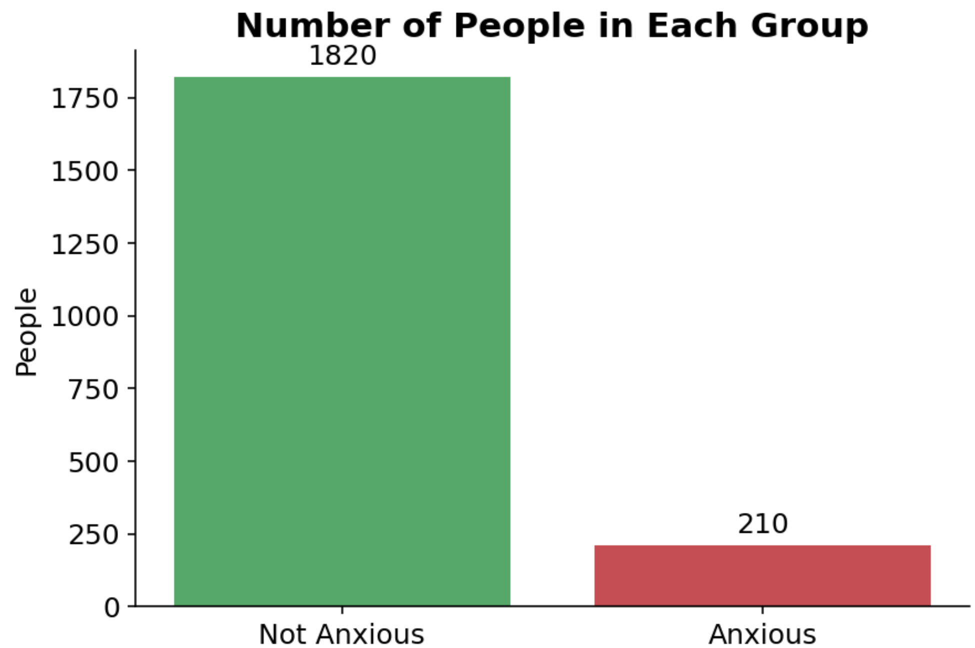
- Fixed contradictions: if someone had attended therapy sessions but their therapy history was blank, we filled that gap logically instead of leaving it empty.
- Removed impossible values: negative numbers (e.g. negative caffeine) were treated as missing rather than kept as errors.
- Handled extreme outliers: a few wildly unrealistic values were smoothed out so one strange entry couldn't distort a whole chart.
- Filled in the blanks sensibly: missing numbers were filled using typical values, and we kept a hidden marker noting which ones were originally missing.
- Grouped numbers into easy labels ("binning"): for example, sleep hours became "Sleep Deprived", "Insufficient", "Recommended", or "Oversleeping", and jobs were grouped into Academic, Healthcare, Professional, Creative, Technical, and Other.
- Saved the result as a single clean file (cleaned_data.csv) that every chart in this report is built from.

WHY THIS MATTERS

Think of it like washing and chopping ingredients before cooking. None of the insights in the rest of this report would be trustworthy without this step. Once the data was clean, we explored it in two ways: category-style charts (groups and labels) and number-style charts (distributions and relationships). The following pages walk through the most useful charts one at a time.

Who in Our Data Feels Anxious?





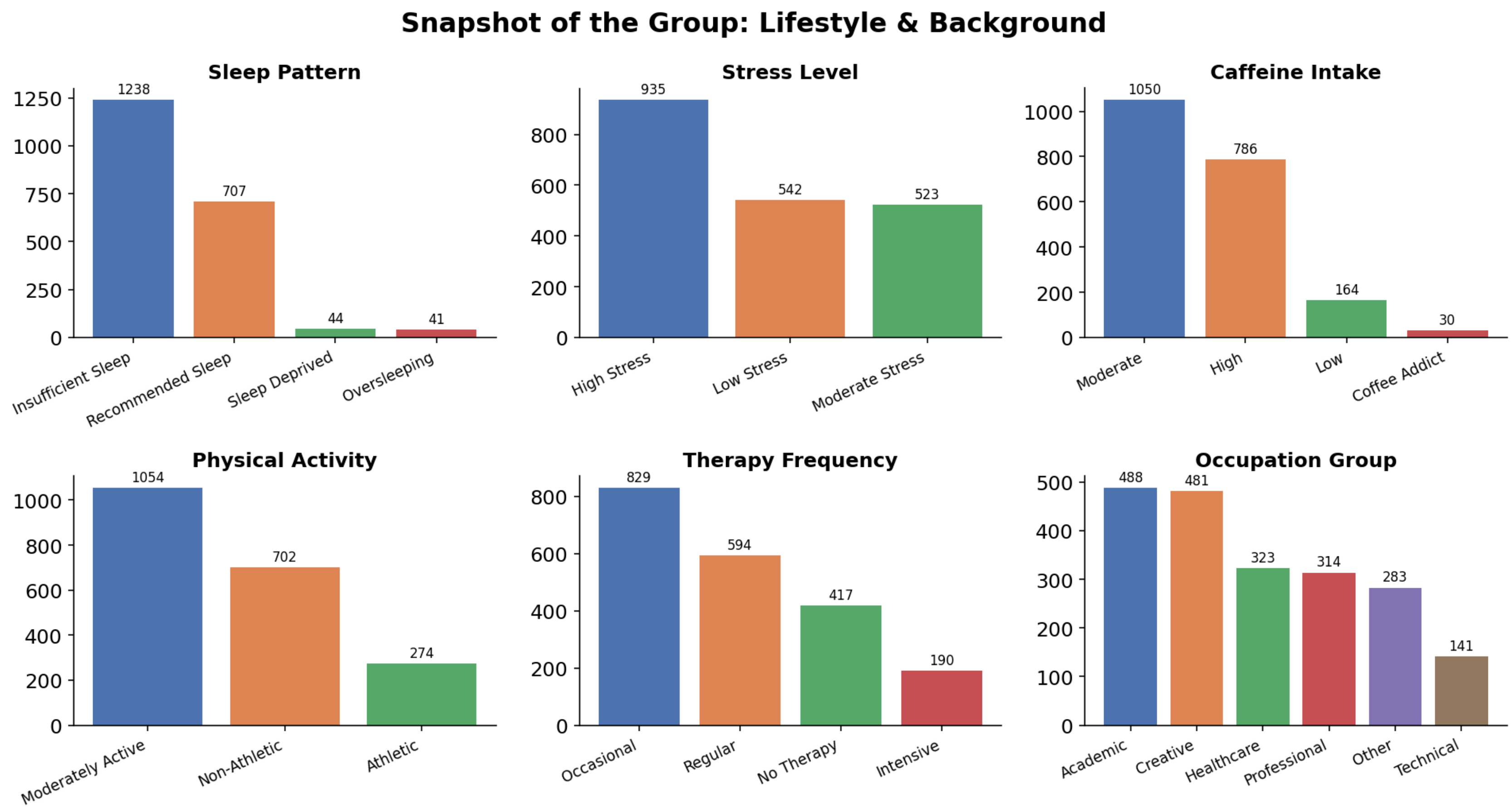
WHAT THIS CHART SHOWS

These two charts answer the most basic question first: out of everyone in the dataset, how many actually report feeling anxious? The donut on the left shows the split as a percentage, and the bars on the right show the actual head-count. “Anxious” means the person was flagged as anxious in the data; “Not Anxious” is everyone else.

CONCLUSION

Most people in the dataset are not anxious — only about 1 in 10 (around 210 of 2,030 people) are. This is important context: because anxious people are the smaller group, the rest of the report focuses on what makes that group different from everyone else.

A Snapshot of the Whole Group



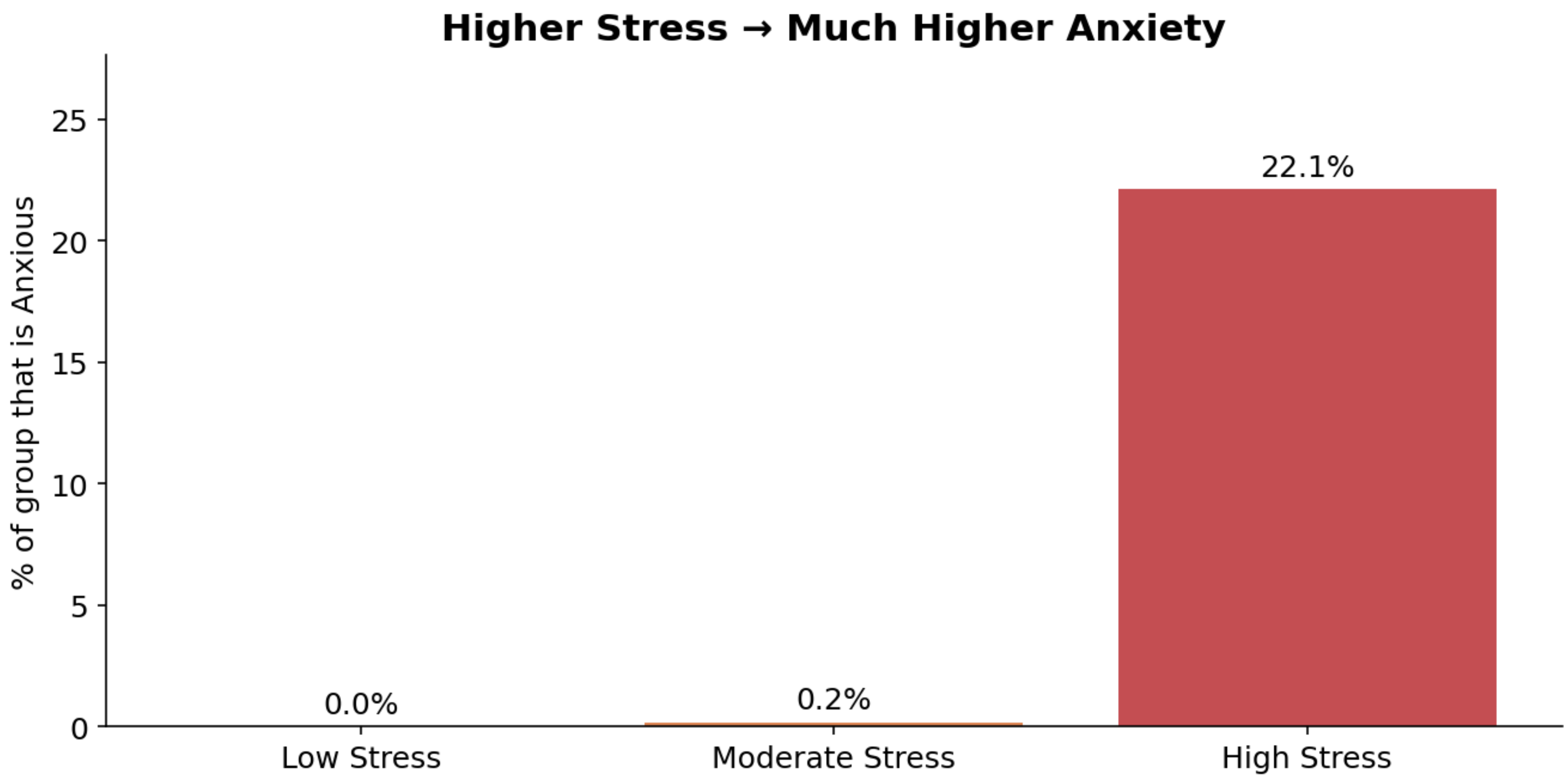
WHAT THESE CHARTS SHOW

Before comparing groups, it helps to know who is in the data. Each small bar chart counts how many people fall into each category for one topic — sleep pattern, stress level, caffeine intake, physical activity, how often they attend therapy, and what kind of job they have. Taller bars mean more people.

CONCLUSION

The typical person here sleeps less than recommended, drinks a moderate amount of caffeine, and is moderately active. Notably, a large share report high stress and insufficient sleep — two things we will see are closely tied to anxiety on the next pages.

Stress Level vs. Anxiety



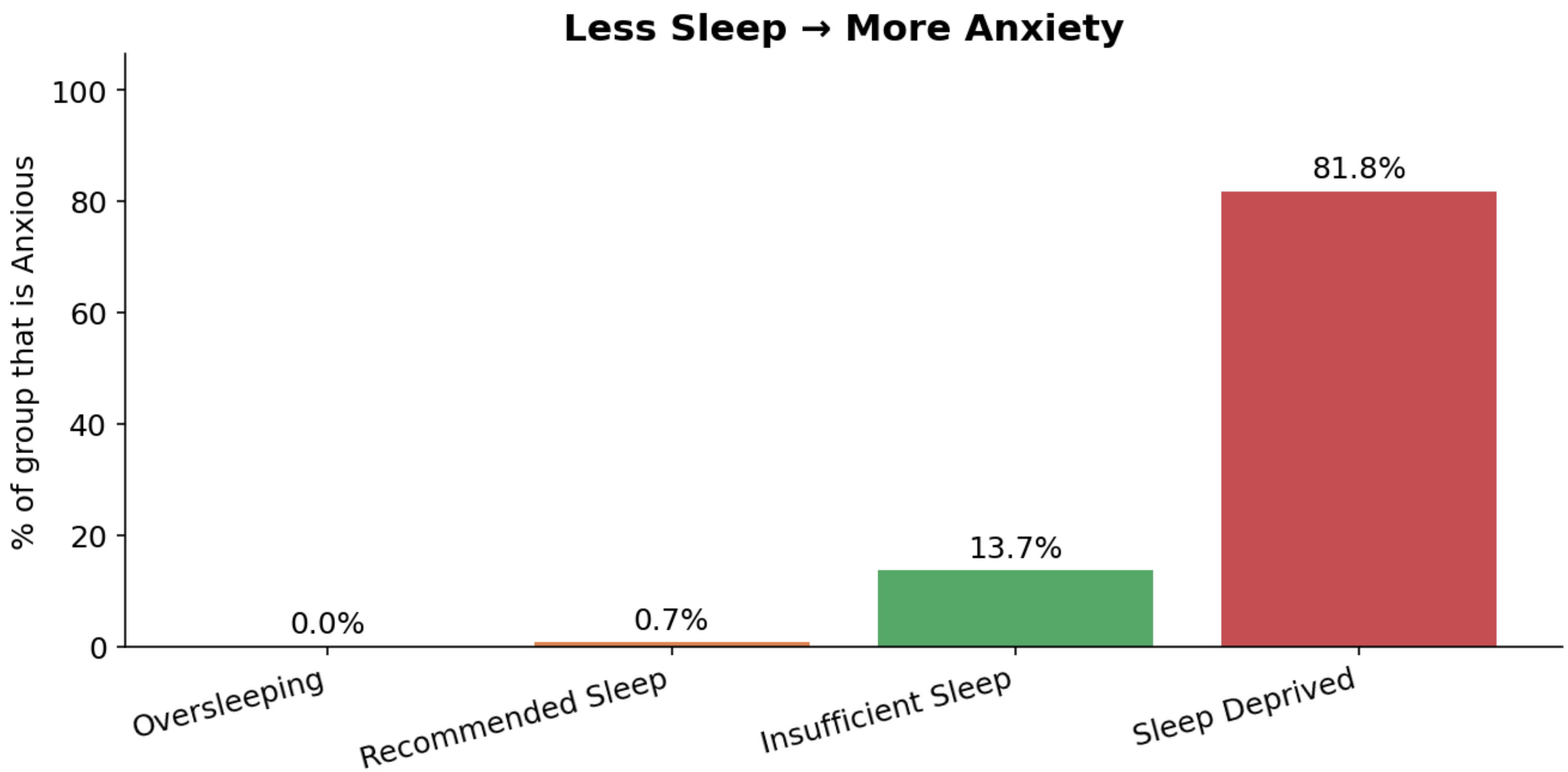
WHAT THIS CHART SHOWS

Here we split people by their stress level (Low, Moderate, High) and, for each group, calculate the share who are anxious. A taller bar means anxiety is more common in that group. This isolates the single strongest categorical signal we found.

CONCLUSION

Anxiety is extremely concentrated in the high-stress group and almost absent among low- and moderate-stress people. Stress level is by far the clearest warning sign of anxiety in this data.

Sleep vs. Anxiety



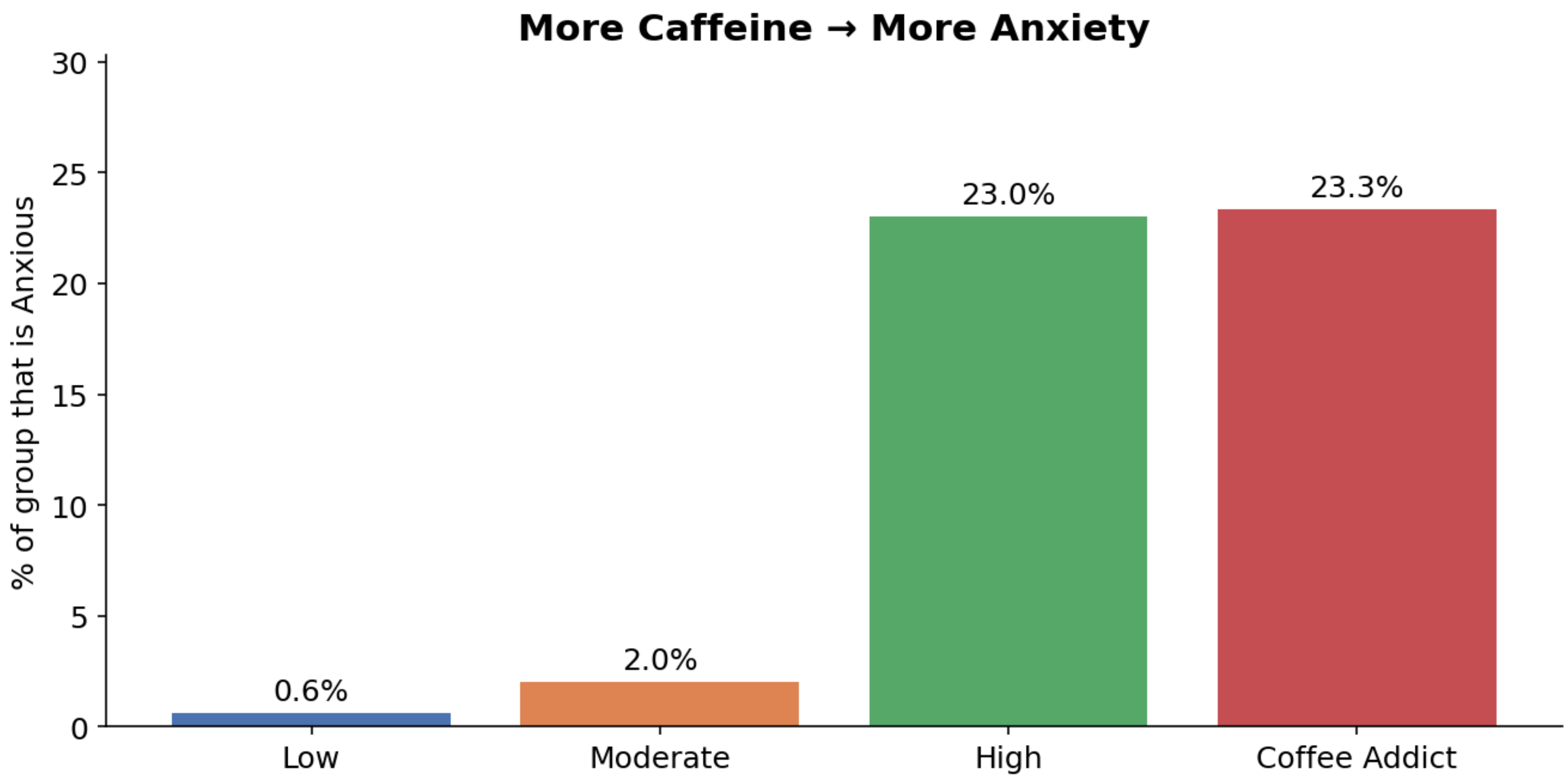
WHAT THIS CHART SHOWS

The same idea applied to sleep. People are grouped by sleep pattern, from oversleeping through to sleep-deprived, and each bar shows the percentage in that group who are anxious. Reading left to right, sleep gets worse.

CONCLUSION

As sleep gets worse, anxiety rises sharply. The sleep-deprived group has a dramatically higher anxiety rate than people who get recommended sleep, making poor sleep one of the strongest lifestyle red flags.

Caffeine vs. Anxiety



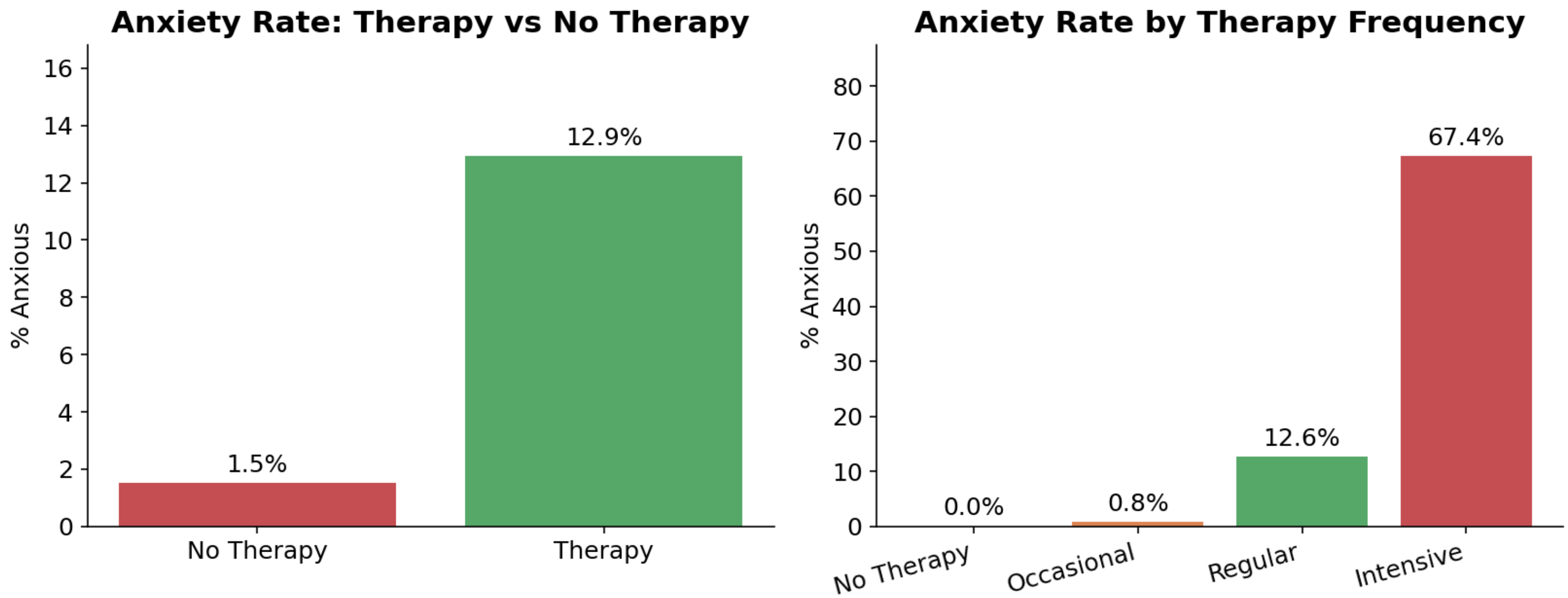
WHAT THIS CHART SHOWS

People are grouped by how much caffeine they consume, from Low up to "Coffee Addict." Each bar shows the percentage in that group who are anxious, so we can see whether heavier caffeine use comes with more anxiety.

CONCLUSION

Anxiety tends to climb with caffeine intake. The heaviest caffeine users show the highest anxiety rates, suggesting caffeine is a meaningful — though smaller — contributor compared with stress and sleep.

Does Therapy Help?



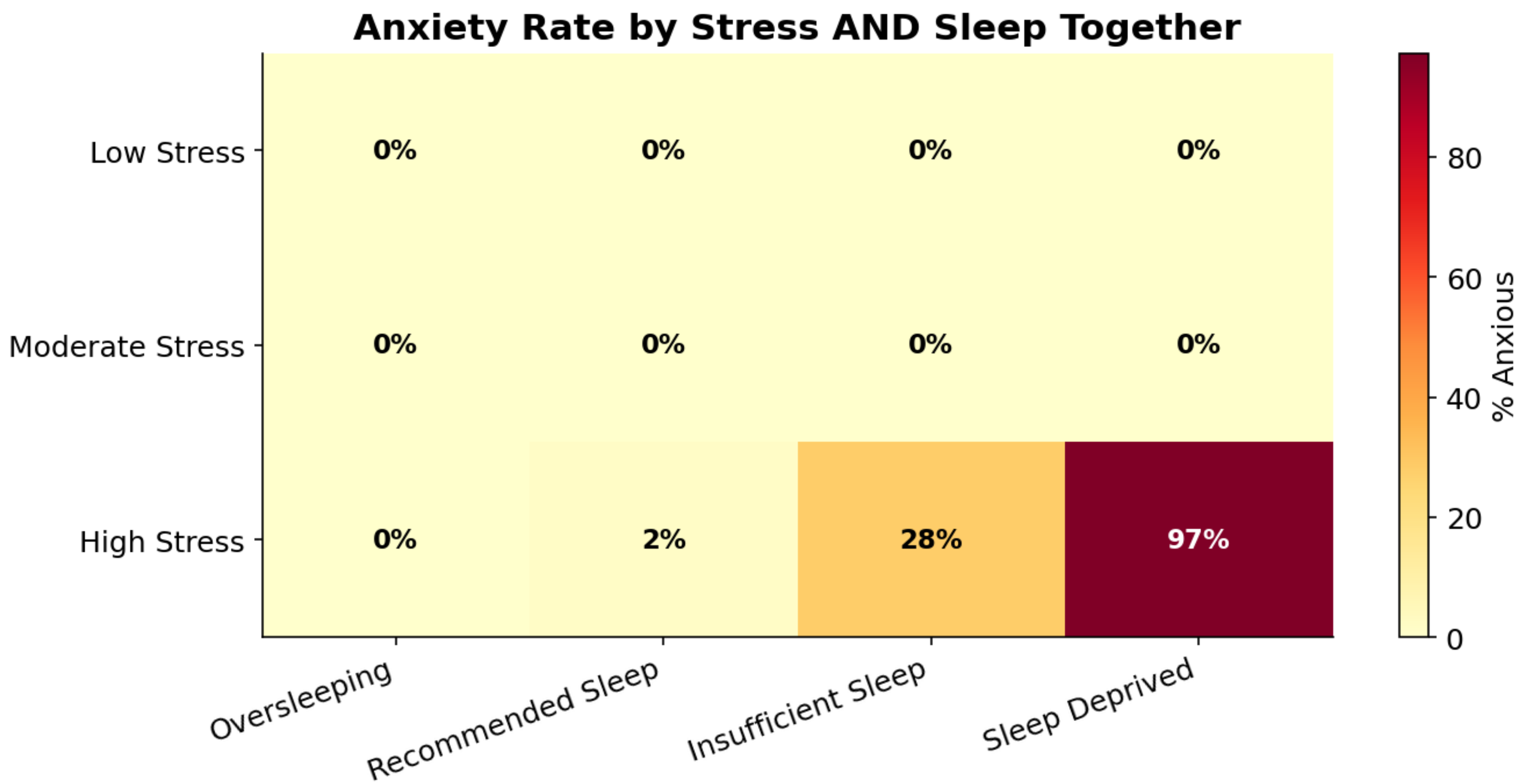
WHAT THIS CHART SHOWS

A natural question: are people who attend therapy less anxious? The left chart compares people who have any therapy against those who have none. The right chart breaks therapy down by how often people go, from none up to intensive. Each bar is the share of that group who are anxious.

CONCLUSION

This is a place to be careful. People in therapy are not automatically less anxious in the data — often the opposite — most likely because anxious people are the ones who seek therapy in the first place. The charts show who goes to therapy, not proof of whether it works. A proper before-and-after study would be needed to judge its effect.

Stress and Sleep Together: The Danger Zone



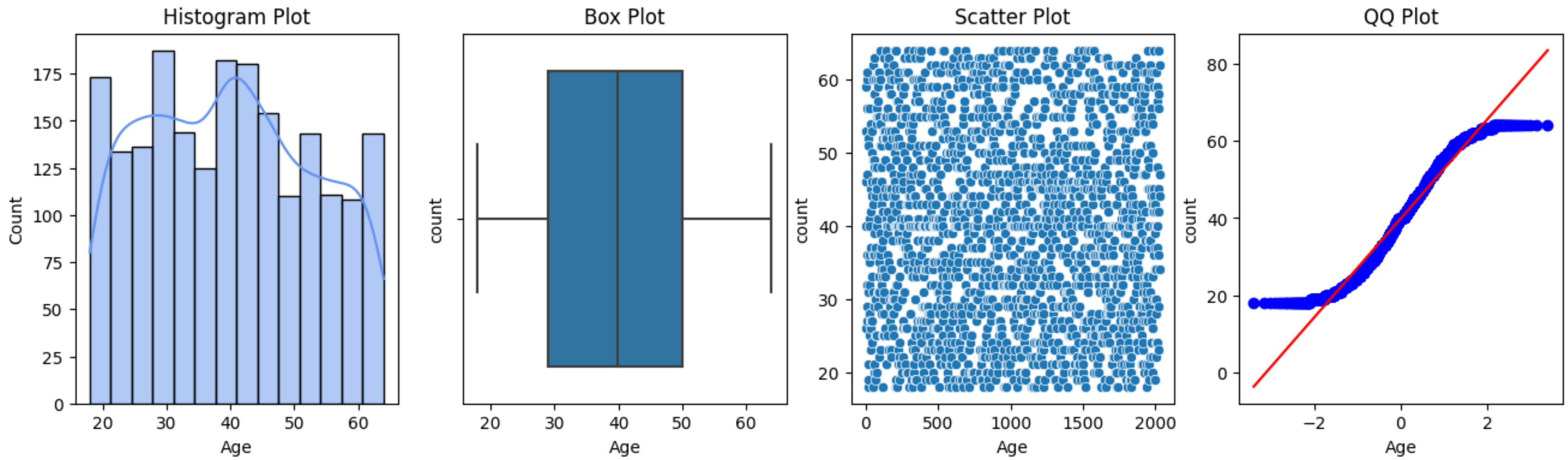
WHAT THIS CHART SHOWS

This grid combines the two strongest factors at once. Every box is a combination of a stress level (rows) and a sleep pattern (columns), and the number inside is the percentage of that exact combination who are anxious. Darker red means a higher anxiety rate.

CONCLUSION

The two factors multiply each other. People who are both highly stressed and sleep-deprived are anxious almost without exception, while almost no one with low or moderate stress is anxious regardless of sleep. The combination of high stress and poor sleep is the clearest “danger zone” in the entire dataset.

Reading a Number Column: The Four-Panel View (Age)



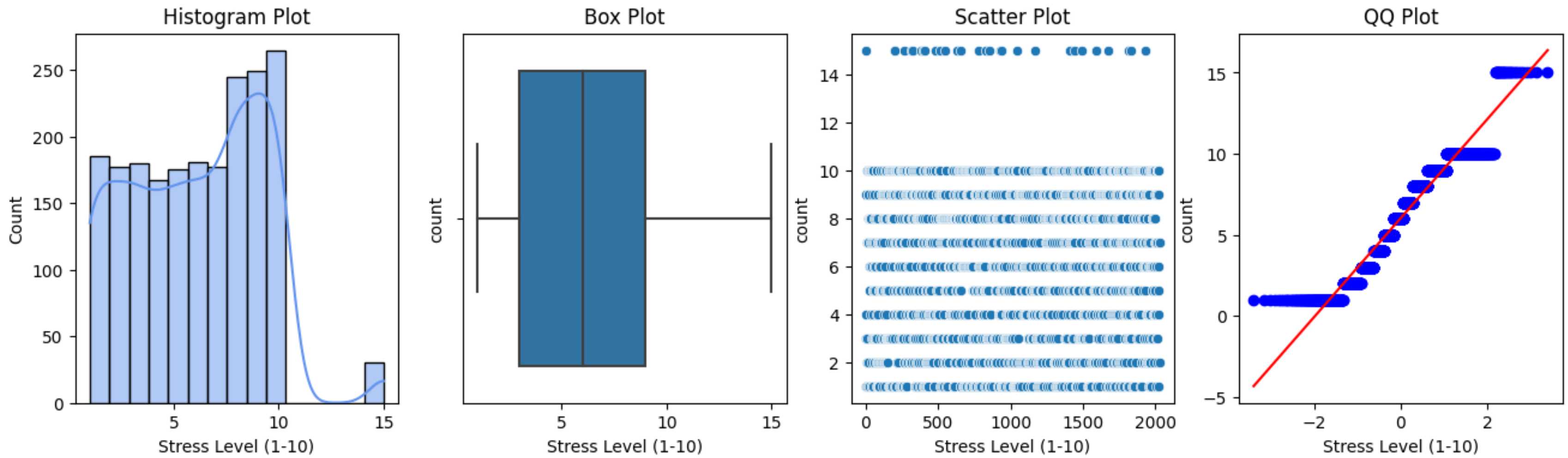
HOW TO READ THESE FOUR PANELS

For every numeric column we drew the same four pictures, using Age as the example here. The Histogram (far left) shows how many people fall at each value. The Box Plot shows the middle of the data and its spread. The Scatter Plot shows each individual person as a dot. The QQ Plot (far right) checks whether the data follows a neat bell-curve shape — dots on the red line means yes, dots curving away means no.

CONCLUSION

Ages are spread fairly evenly across adulthood with no single dominant age, and the QQ plot shows the spread is broad rather than a perfect bell curve. The same four-panel layout is used for the other measurements that follow.

How Stress Levels Are Distributed



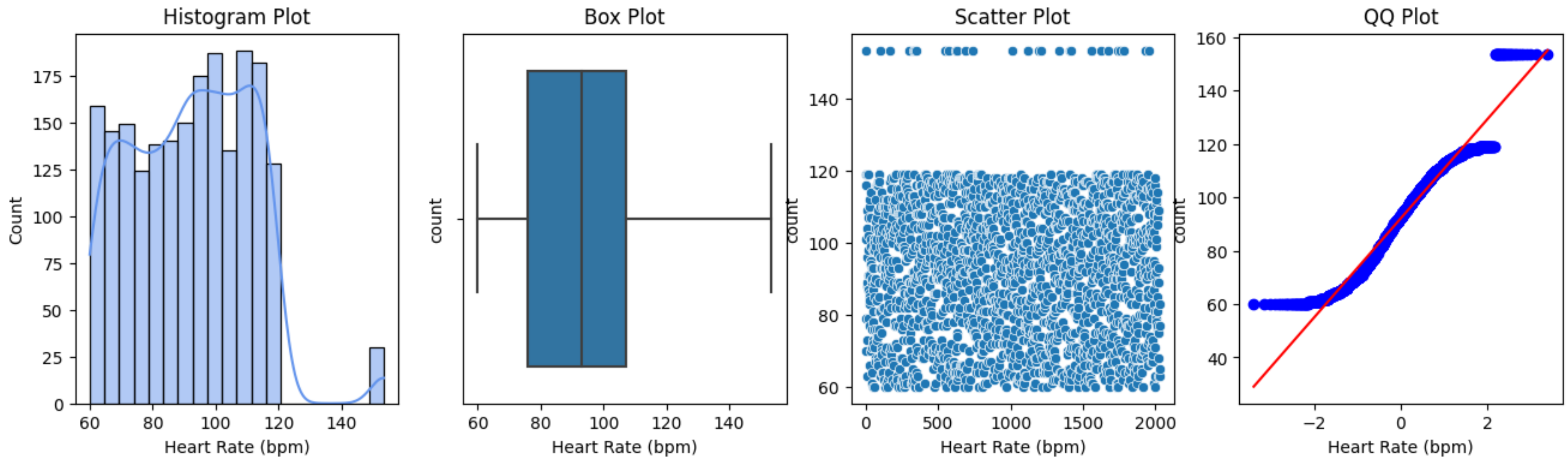
WHAT THIS CHART SHOWS

The same four-panel view, now for the raw stress score (1 to 10). The histogram shows how common each stress score is, the box and scatter show the spread, and the QQ plot checks the shape.

CONCLUSION

Stress scores cover the full range rather than clustering at “calm.” A substantial chunk of people sit at the higher end of the stress scale — consistent with the earlier finding that high stress is widespread and strongly tied to anxiety.

How Heart Rate Is Distributed



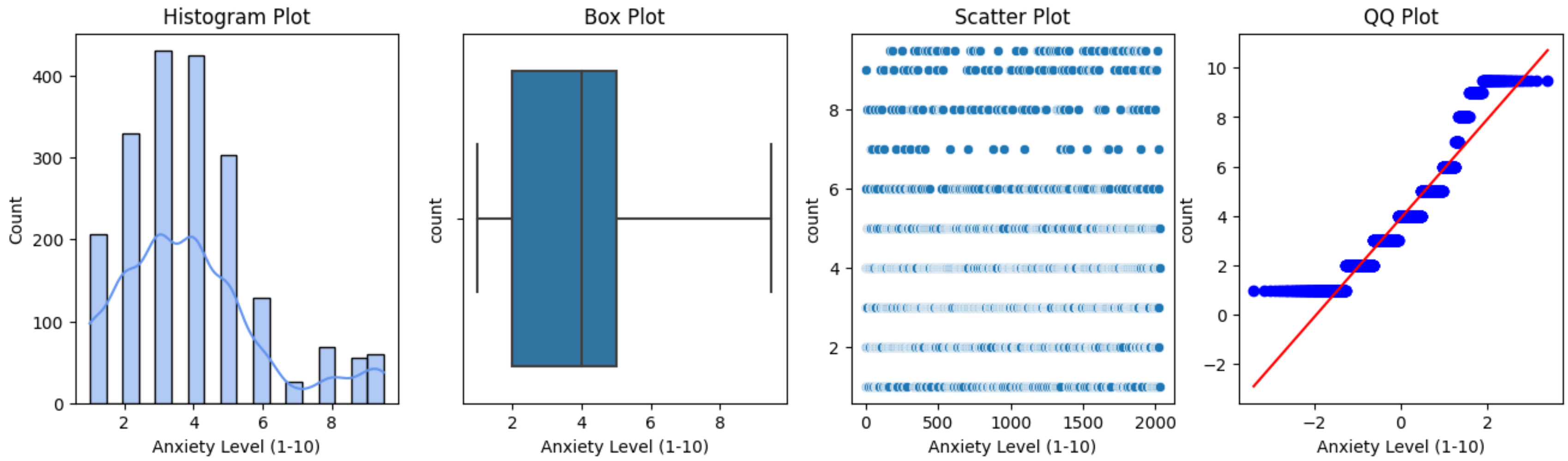
WHAT THIS CHART SHOWS

Heart rate (beats per minute) shown in the four-panel view. Because heart rate is a physical body signal, its shape tells us whether most people sit in a normal range or whether there are many unusually high readings.

CONCLUSION

Most heart-rate readings cluster in a normal, healthy band with a fairly balanced bell-like shape. There is no large group of extreme readings, which means heart rate on its own is a weaker anxiety signal than stress or sleep.

How the Anxiety Score Is Distributed



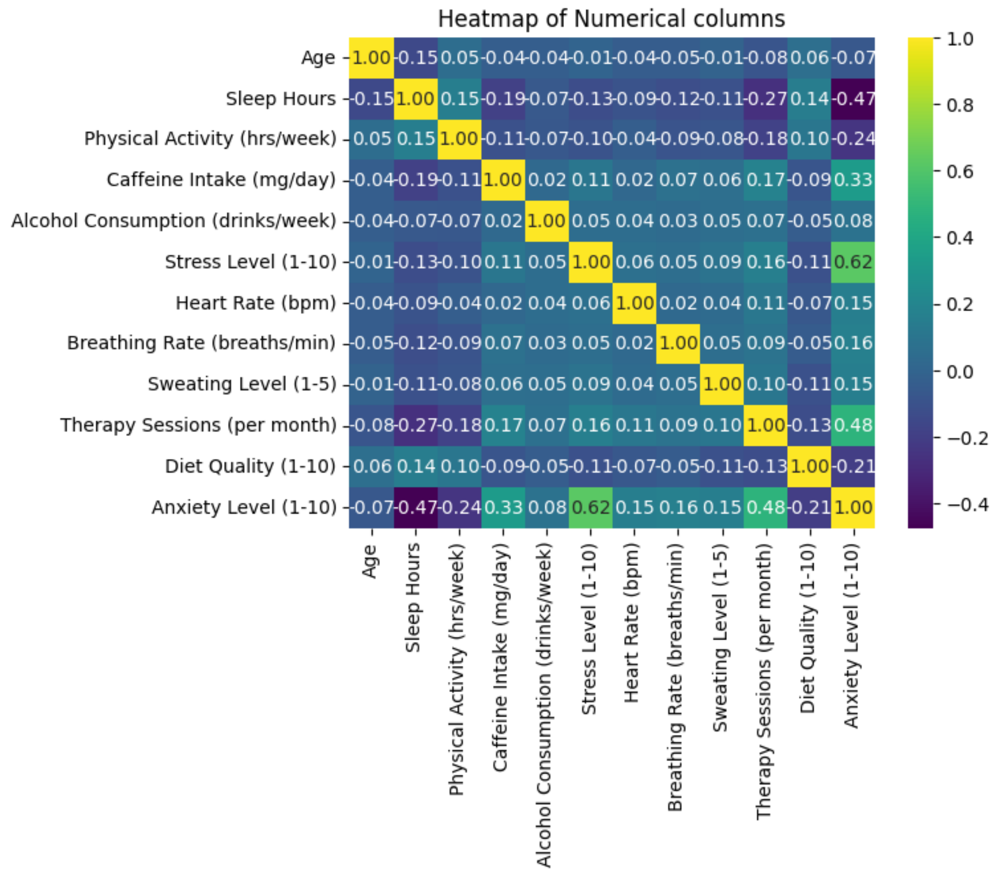
WHAT THIS CHART SHOWS

This is the four-panel view of the underlying anxiety score itself (1 to 10), before it was simplified into the Anxious / Not-Anxious label used earlier. It shows how anxiety is spread across everyone.

CONCLUSION

Most people sit at the lower end of the anxiety scale, with a smaller tail of people reporting much higher scores. This matches the opening chart: severe anxiety is real but affects a minority, which is exactly why spotting its warning signs matters.

Which Measurements Move Together?



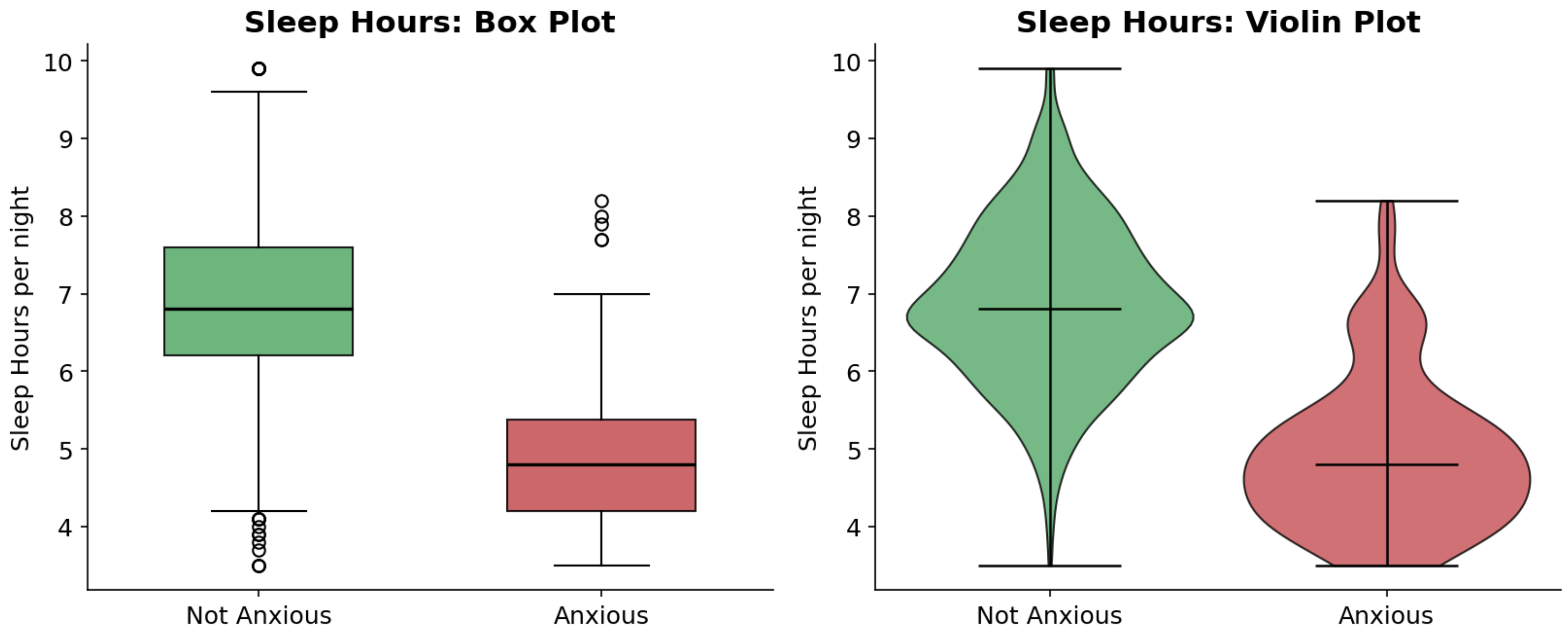
WHAT THIS CHART SHOWS

This is a correlation heatmap. Every square compares two measurements and shows how strongly they rise and fall together, from -1 to +1. Bright yellow (close to +1) means “when one goes up, so does the other.” Dark blue (negative) means “when one goes up, the other goes down.” Read the bottom “Anxiety Level” row to see what is linked to anxiety.

CONCLUSION

Anxiety rises most with Stress (+0.62) and Therapy Sessions (+0.48), and falls most as Sleep improves (−0.47). In plain terms: more stress, less sleep, and more therapy-seeking all travel alongside higher anxiety — the same story the category charts told, now confirmed with numbers.

Comparing Sleep Between Anxious and Calm People



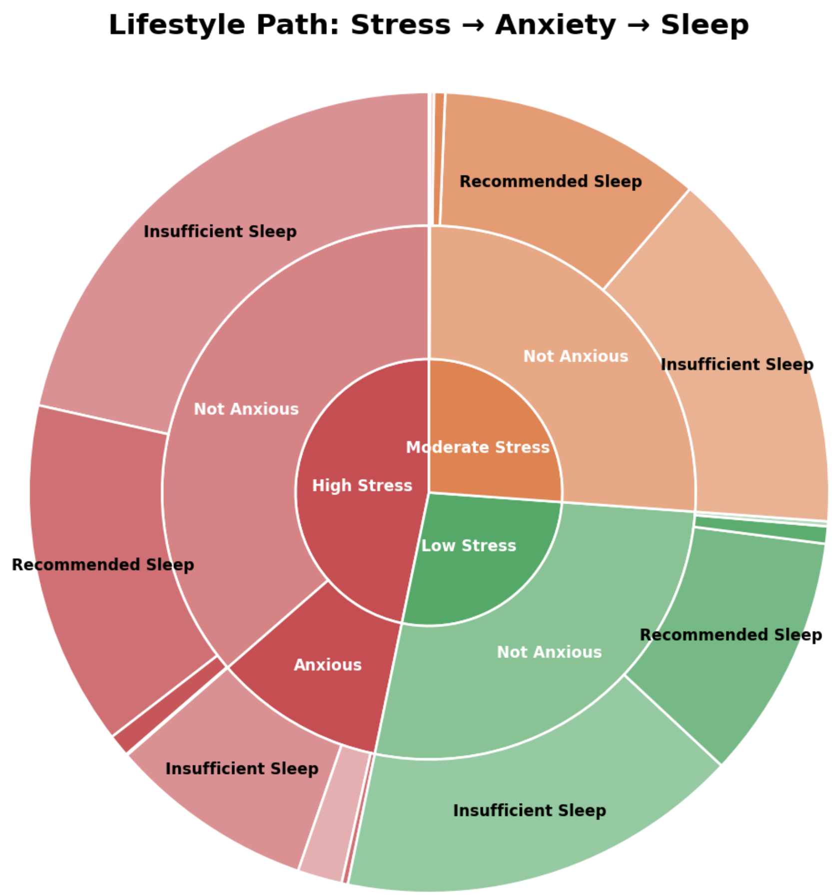
WHAT BOX AND VIOLIN PLOTS SHOW

Box and violin plots are the best way to compare a number between two groups. We split everyone into Anxious and Not-Anxious, then look at how many hours they sleep. In the Box Plot (left), the line inside each box is the typical (middle) value and the box covers most people. The Violin Plot (right) adds shape — the wider the bulge, the more people sleep that amount.

CONCLUSION

The two groups barely overlap. Calm people typically sleep close to seven hours a night, while anxious people typically sleep under five. This is one of the sharpest contrasts in the whole report and confirms poor sleep as a leading companion of anxiety.

The Full Picture in One Circle: Stress → Anxiety → Sleep



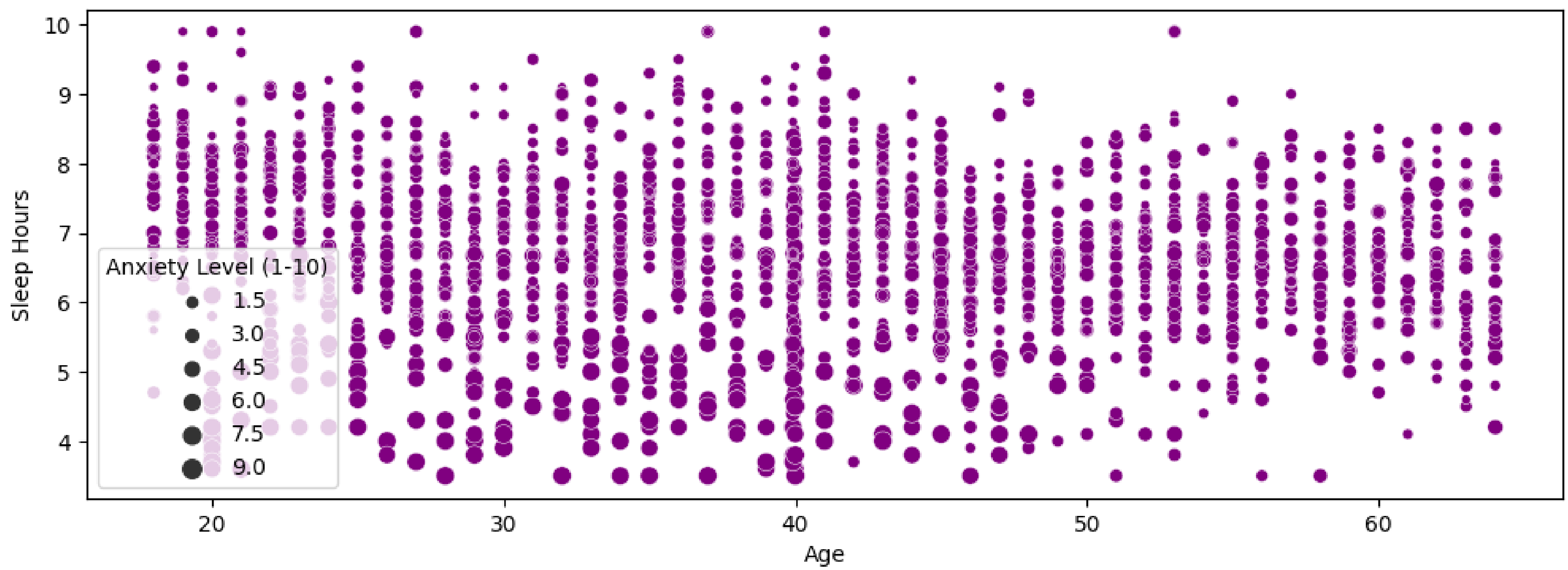
HOW TO READ THIS SUNBURST

A sunburst chart shows layers from the inside out, like tree rings. The inner circle splits everyone by stress level (red = high, orange = moderate, green = low). The middle ring splits each of those into Anxious vs Not Anxious. The outer ring then splits those by sleep pattern. The bigger a slice, the more people it contains — so you can follow any path from the centre outward to a specific kind of person.

CONCLUSION

Follow the red (High Stress) wedge outward: it is the only place where a large “Anxious” band appears, and within it the anxious people are almost all sleep-deprived or sleeping too little. The green (Low Stress) and orange (Moderate) wedges are overwhelmingly “Not Anxious.” In a single picture, this ties the whole report together: anxiety lives where high stress and poor sleep meet.

Three Things at Once: Age, Sleep and Anxiety



WHAT THIS CHART SHOWS

This scatter plot shows three pieces of information together. Each dot is a person: their position left-to-right is their Age, up-and-down is their Sleep Hours, and the size of the dot is their Anxiety Level — bigger dots mean more anxious.

CONCLUSION

The biggest dots (most anxious people) cluster toward the bottom of the chart, where sleep hours are lowest, and appear at every age. This visually reinforces the headline of the whole report: too little sleep travels with higher anxiety, no matter how old someone is.

In Summary: What the Data Told Us

Putting all the charts together, a consistent and simple story emerges from the data.

THE MAIN TAKEAWAYS

- Anxiety is a minority experience here — about 1 in 10 people — but it has clear, repeating warning signs.
- High stress is the single strongest signal linked to anxiety.
- Poor sleep is the second strongest — the worse the sleep, the higher the anxiety.
- High stress AND poor sleep together form a “danger zone” where almost everyone is anxious.
- Heavier caffeine use lines up with somewhat more anxiety, while heart rate and gender show little direct link.
- People in therapy aren’t less anxious in this snapshot — likely because anxious people seek therapy — so this data can’t judge whether therapy works.

AN IMPORTANT REMINDER

These charts show patterns that travel together; they do not prove that one thing causes another. “Linked to” is not the same as “causes.” Still, the patterns are strong and consistent, and they point clearly to stress management and healthy sleep as the most promising places to focus.

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